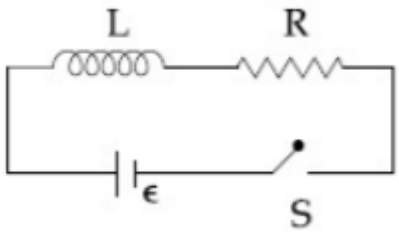
8th January shift-2

Physics

Q. 1



As shown in the figure, a battery of emf ϵ is connected to an inductor L and resistance R in series. The switch is closed at t = 0. The total charge that flows from the battery, between t = 0 and t = t_c (t_c is the time constant of the circuit) is :

Option 1:
$\frac{\epsilon L}{R^2}\left(1-\frac{1}{e}\right)$

Option 2:
$\frac{\epsilon L}{R^2}$

Option 3:
$\frac{\epsilon R}{eL^2}$

Option 4:
$\frac{\epsilon L}{eR^2}$

Correct Answer:
$\frac{\epsilon L}{eR^2}$

Solution:

As current at any time t is given as

$$I = I_0\left(1-e^{\frac{-t}{\tau}}\right) = \frac{\epsilon}{R}\left(1-e^{\frac{-t}{\tau}}\right)$$

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So $q = \int_0^{T_C} idt$

So integrating this will give total charge

$$q = \frac{\varepsilon}{R}\left[t - \frac{e^{-t/\tau}}{\frac{-1}{\tau}}\right]_0^{\tau}; \quad = \frac{\varepsilon}{R}\left[\tau + \tau e^{-1} - \tau\right]$$

$$q = \frac{\varepsilon}{R} \times \frac{1}{e} \times \frac{L}{R} \quad ; \quad q = \frac{\varepsilon L}{R^2 e}$$

Hence the correct option is (4).

Q. 2 A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the pendulum is 25.0 cm and a stop watch with 1 s resolution measures the time taken for 40 oscillations to be 50 s. The accuracy in g is :

Option 1:
2.40%

Option 2:
5.40%

Option 3:
4.40%

Option 4:
3.40%

Correct Answer:
4.40%

Solution:

As

- **The time period of oscillation of simple pendulum (T)-**

$$T = 2\pi\sqrt{\frac{l}{g}}$$

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Q. 12 The length of the perpendicular from the origin, on the normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at the point (2,2) is ;

Option 1:
2

Option 2:
$2\sqrt{2}$

Option 3:
$4\sqrt{2}$

Option 4:
$\sqrt{2}$

Correct Answer:
$2\sqrt{2}$

Solution:

Line parallel and perpendicular to a given line -

Line parallel and perpendicular to a given line

The equation of the line parallel to ax + by + c = 0 is given as ax + by + λ = 0, where λ is some constant.

Equation of the given line is ax + by + c = 0

Its slope is (-a/b)

So, any equation of line parallel to ax + by + c = 0 is

$$y = \left(-\frac{a}{b}\right)x + c_1$$
$$ax + by - bc_1 = 0$$
$$ax + by + \lambda = 0$$

The equation of the line perpendicular to ax + by + c = 0 is given as bx - ay + λ = 0, where λ is some constant.

Equation of the given line is ax + by + c = 0

Its slope is (-a/b)

Slope of perpendicular line will be (b/a)

So, any equation of line perpendicular to ax + by + c = 0 is

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$$y = \left(\frac{b}{a}\right)x + c_1$$
$$bx - by - ac_1 = 0$$
$$bx - ay + \lambda = 0$$

-

Homogeneous Equations in Two Variables -

Homogeneous Equations in Two Variables

Homogeneous equations are those equations where each term has the same degree.

The equation $ax^2 + 2hxy + by^2 = 0$ is a homogeneous equation of second degree, it represents two straight lines through the origin.

The given equation is :

$$ax^2 + 2hxy + by^2 = 0 \qquad \ldots(i)$$

Divide the equation by x^2

$$\Rightarrow \qquad a + 2h\left(\frac{y}{x}\right) + b\left(\frac{y^2}{x^2}\right) = 0$$

now put $\frac{y}{x} = m$

$$\Rightarrow \qquad a + 2h(m) + b(m^2) = 0 \qquad \ldots(ii)$$

If m_1 and m_2 are roots, then

$$m_1 + m_2 = -\frac{2h}{b}$$

amd, $\qquad m_1 m_2 = \frac{b}{a}$

Thus, $y = m_1 x$ and $y = m_2 x$ are two straight lines given by Eq. (i)

Also, from the eq (iii)

$$m = \frac{-2h \pm \sqrt{(h^2 - ab)}}{2b} = \frac{y}{x}$$

$$\therefore \qquad by = \left\{-h + \sqrt{(h^2 - ab)}\right\} x$$

and $\qquad by = \left\{-h - \sqrt{(h^2 - ab)}\right\} x$

are two lines represented by Eq. (i)

-

Distance of a Point From a Line -

Distance of a point from a line

Perpendicular length from a point (x1,y1) to the line L : Ax + By + C = 0 is

$$\frac{|\mathbf{Ax_1} + +\mathbf{By_1} + \mathbf{C}|}{\sqrt{\mathbf{A^2} + \mathbf{B^2}}}$$

-

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$x^2 + 2xy - 3y^2 = 0$
$x^2 + 3xy - xy - 3y^2 = 0$
$(x - y)(x + 3y) = 0$
$x - y = 0 \quad x + 3y = 0$

(2, 2) satisfy x – y = 0

Normal

x + y = λ= 4 Hence

the perpendicular distance from the origin $= \left|\frac{0 + 0 - 4}{\sqrt{2}}\right| = 2\sqrt{2}$

Correct Option (2)

Q. 13 Which of the following statements is a tautology ?

Option 1:
$\sim (p \wedge \sim q) \rightarrow p \vee q$

Option 2:
$\sim (p \vee \sim q) \rightarrow p \wedge q$

Option 3:
$p \vee (\sim q) \rightarrow p \wedge q$

Option 4:
$\sim (p \vee \sim q) \rightarrow p \vee q$

Correct Answer:
$\sim (p \vee \sim q) \rightarrow p \vee q$

Solution:

Tautology And Contradiction -

Tautology

A compound statement is called **tautology** if it is always true for all possible truth values of its component statement.

For example, (p ⇒ q) ∨ (q ⇒ p)

Contradiction (fallacy)

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Q. 9 The electronic configurations of bivalent europium and trivalent cerium are :

(atomic number : Xe = 54, Ce = 58, Eu = 63)

Option 1:
$[Xe]4f^7 \ and \ [Xe]4f^1$

Option 2:
$[Xe]4f^7 6s^2 \ and \ [Xe]4f^2 6s^2$

Option 3:
$[Xe]4f^2 \ and \ [Xe]4f^7$

Option 4:
$[Xe]4f^4 \ and \ [Xe]4f^9$

Correct Answer:
$[Xe]4f^7 \ and \ [Xe]4f^1$

Solution:
The electronic configurations are given as

Eu^{2+} : $[Xe]4f^7$
Ce^{3+} : $[Xe]4f^1$

Hence, the option number (1) is correct.

Q. 10 B has smaller first ionization enthalpy than Be. Consider the following statements :

(I) It is easier to remove 2p electron than 2s electron.

(II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be.

(III) 2s electron has more penetration power than 2p electron

(IV) atomic radius of B is more than Be

(atomic number B = 5, Be = 4)

The correct statements are :

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